

Sensitivity-Engine Derivative Surrogates: Greek- and Structure-Consistent Supervision*

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Abstract

This paper develops a derivative-surrogate framework in which prices, first-order Greeks, selected second-order Greeks, and probabilistic decomposition blocks are trained jointly under explicit internal-consistency restrictions. The purpose is not to replace established pricing theory or classical numerical methods, but to transfer the structure of a sensitivity engine into the statistical objective and architecture of a surrogate.

The empirical implementation uses public S&P 500, Treasury, and volatility-index series to construct date-causally aligned daily states from December 4, 2007 through March 18, 2026. An analytic Black–Scholes–Merton teacher is evaluated on nine maturities and fifteen log-moneyness points per date, producing 621,000 labeled observations. Four models are compared across five matched seeds: a scalar price-potential model; a direct price-and-Greek model; a potential-consistent model that ties direct Greek heads to automatic derivatives of a model-specific normalized-price potential; and a decomposition-consistent model with supervised asset-leg and strike-leg probability heads.

The completed extended-horizon run uses a 960-epoch safety cap for Model A and a 320-epoch cap for Models B–D; all twenty fits terminate through the common early-stopping rule without reaching their applicable caps. Three principal findings emerge. First, direct Greek supervision lowers global Gamma, Vega, Vanna, and Volga RMSE by approximately 70%, 42%, 71%, and 76%, respectively, relative to the price-only Model A, but Delta worsens directionally and price RMSE rises materially. Second, potential consistency reduces the five head–potential gap RMSEs by approximately 81%–97% and lowers price RMSE relative to the unconstrained direct-head model by about 50% globally and 68% on the high-curvature slice; it does not improve every standalone Greek RMSE. Third, the two-leg model learns the probability decomposition with an auxiliary normalized-price gap of approximately 3.4×10^{-4} between its direct and reconstructed price heads and significantly improves global price RMSE relative to the potential-consistent model, but the scalar-potential model retains the lowest mean global price and Delta RMSE. The two-leg model also has significantly larger global Delta, Vega, and Volga errors than the potential-consistent model. Five-day hedging statistics remain economically similar across models and do not establish structural dominance. The evidence therefore supports Greek supervision, potential consistency, and decomposition fidelity as distinct objectives rather than a monotone hierarchy in which additional structure improves every task.

*This paper builds on *BSM Sensitivity Engine* (BSE) [17], *Distributional Sensitivity Engine* (DSE) [18], and *Numeraire-Based Distributional Sensitivity Engine* (NDSE) [19]. The empirical implementation is available at https://github.com/DavidHShen/Sensitivity-Engine_Derivative_Surrogates.

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JEL: G12, G13, C45, C63

1 Introduction

Machine-learning surrogates can deliver low-latency derivative valuation across large state spaces, but price accuracy alone is not sufficient for risk-sensitive applications. Neural-network approximation of derivative prices and hedges dates at least to Hutchinson, Lo, and Poggio [4], while modern work uses deep networks as offline surrogates for pricing and calibration in computationally demanding volatility models [10]. Hedging, scenario analysis, calibration, and aggregation require stable local sensitivities. A surrogate may interpolate prices adequately while producing a Delta, Gamma, or cross-Greek field that is incompatible with any coherent pricing function. This distinction becomes important whenever the surrogate is used as a risk engine rather than as a scalar interpolation device.

Training on derivative labels is related to Sobolev training, which augments function-value supervision with target derivatives [6], and to Differential Machine Learning, which develops this principle specifically for derivative pricing and risk approximation [7]. The present paper differs by separating direct Greek accuracy from agreement with a common normalized pricing potential and from fidelity to a supervised two-leg probabilistic decomposition.

The earlier sensitivity-engine sequence suggests a direct response. *BSM Sensitivity Engine* (BSE) derives implementation-oriented first- and mixed second-order Black–Scholes–Merton sensitivities from a compact generator representation [1, 2, 17]. *Distributional Sensitivity Engine* (DSE) rewrites derivative values and sensitivities through two-measure digital weights and conditional-law blocks under deterministic discounting [18]. *Numeraire-Based Distributional Sensitivity Engine* (NDSE) preserves the two-leg structure under stochastic discounting by matching each leg to an appropriate numeraire and measure [3, 19]. These engines provide not only labels but also algebraic relations among prices, Greeks, primitive coefficients, and probabilistic weights.

The present paper transfers those relations into trainable restrictions. Three questions are central.

- Q1.** Does direct supervision on selected Greeks improve risk fidelity relative to price-only learning?
- Q2.** Does agreement between direct heads and the automatic derivatives of a common pricing potential reduce internal inconsistency?
- Q3.** Does explicit supervision of the two-leg probabilistic representation improve price, sensitivity, or hedging performance beyond a potential-consistent multi-output network?

The contribution is intentionally narrower than a claim that every additional structural restriction must improve every metric. A structure-consistent model can improve integrability or decomposition fidelity while accepting a tradeoff in one standalone RMSE. The relevant comparison is therefore multi-dimensional: price accuracy, sensitivity accuracy, consistency residuals, probability-head error, and hedge-error behavior are reported separately.

The empirical design also separates training from local diagnosis. A short-maturity near-the-money slice is retained because curvature is concentrated there, but that slice does not alter sampling,

loss weights, or model selection. It is an ex-post high-curvature diagnostic, not a distinct training intervention. All four models use the same chronological sample, optimizer, validation score, early-stopping patience, and model-selection rule. Model A is given a higher safety cap only because its scalar-potential optimization requires more epochs to reach the shared stopping criterion.

The paper makes four contributions. First, it derives a normalized potential representation that links a scalar call-value surrogate to five sensitivity coordinates. Second, it introduces direct head-potential consistency penalties, including mixed-derivative identities. Third, it implements a supervised two-leg representation aligned with DSE. Fourth, it reports a matched multi-seed empirical protocol with date-causal alignment, numerical audits, smooth second-order automatic differentiation, and direct tests of the tradeoffs among pointwise accuracy, structural coherence, and hedging diagnostics.

2 Sensitivity-engine structure

2.1 Analytic teacher labels

Let $C(S, K, r, \tau, \sigma)$ denote a European call value with zero dividend yield. The Black–Scholes–Merton representation is

$$C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \quad d_1 = \frac{\log(S/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}. \quad (1)$$

Here S is spot, K is strike, r is the continuously compounded rate in the analytic model, τ is time to maturity, σ is volatility, and Φ is the standard normal distribution function. The standard normal density is denoted by φ . Throughout the derivations, $S > 0$, $K > 0$, $\tau > 0$, and $\sigma > 0$. The empirical construction of the rate input is stated separately in Section 4.1.

The Greek conventions used in the paper are

$$\Delta := \frac{\partial C}{\partial S}, \quad \Gamma := \frac{\partial^2 C}{\partial S^2}, \quad \text{Vega} := \frac{\partial C}{\partial \sigma}, \quad \text{Vanna} := \frac{\partial^2 C}{\partial S \partial \sigma}, \quad \text{Volga} := \frac{\partial^2 C}{\partial \sigma^2}. \quad (2)$$

Spot derivatives in (2) hold (K, r, τ, σ) fixed, and volatility derivatives hold (S, K, r, τ) fixed.

The analytic teacher supplies

$$\Delta = \Phi(d_1), \quad (3)$$

$$\Gamma = \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}}, \quad (4)$$

$$\frac{\text{Vega}}{S} = \varphi(d_1)\sqrt{\tau}, \quad (5)$$

$$\text{Vanna} = -\frac{\varphi(d_1)d_2}{\sigma}, \quad (6)$$

$$\frac{\text{Volga}}{S} = \frac{\varphi(d_1)\sqrt{\tau}d_1d_2}{\sigma}. \quad (7)$$

The normalized quantities C/S , $S\Gamma$, Vega/S , and Volga/S limit scale differences caused by the level of the index.

2.2 Normalized price potential

Write

$$C(S, K, r, \tau, \sigma) = S c(k, \tau, r, \sigma), \quad k = \log(K/S). \quad (8)$$

For each fixed (τ, r) , assume that c is twice continuously differentiable in (k, σ) on an open set containing the states under consideration. When K is held fixed, $\partial_S k = -1/S$. Differentiation gives

$$\Delta = c - c_k, \quad (9)$$

$$S\Gamma = c_{kk} - c_k, \quad (10)$$

$$\frac{\text{Vega}}{S} = c_\sigma, \quad (11)$$

$$\text{Vanna} = c_\sigma - c_{k\sigma}, \quad (12)$$

$$\frac{\text{Volga}}{S} = c_{\sigma\sigma}. \quad (13)$$

Subscripts denote partial derivatives of the normalized potential c with respect to the indicated raw coordinates.

Proposition 2.1 (Potential-generated consistency). Under the stated regularity assumptions, the five quantities in (13) are generated by one scalar pricing potential, and the mixed partials satisfy $c_{k\sigma} = c_{\sigma k}$.

Proof. Holding (K, r, τ, σ) fixed for the spot derivative,

$$\frac{\partial C}{\partial S} = c + S c_k \frac{\partial k}{\partial S} = c - c_k. \quad (14)$$

Differentiating once more with respect to S yields

$$\Gamma = \frac{\partial}{\partial S}(c - c_k) = (c_k - c_{kk}) \frac{\partial k}{\partial S} = \frac{c_{kk} - c_k}{S}. \quad (15)$$

Holding (S, K, r, τ) fixed for volatility derivatives gives

$$\frac{\partial C}{\partial \sigma} = S c_\sigma, \quad \frac{\partial^2 C}{\partial \sigma^2} = S c_{\sigma\sigma}. \quad (16)$$

Finally,

$$\frac{\partial^2 C}{\partial S \partial \sigma} = \frac{\partial}{\partial S}(S c_\sigma) = c_\sigma + S c_{k\sigma} \frac{\partial k}{\partial S} = c_\sigma - c_{k\sigma}. \quad (17)$$

Equality $c_{k\sigma} = c_{\sigma k}$ follows from Clairaut's theorem.

The proposition gives a sufficient potential-based construction; it is not presented as a converse characterization of every admissible collection of independently specified Greek fields. A direct multi-head network does not satisfy these relations automatically. It can fit each teacher coordinate separately while producing a sensitivity field that is inconsistent with its own price head. The potential-consistency model therefore supervises both teacher error and head-potential disagreement.

2.3 Two-leg probabilistic representation

The normalized call value can also be written as

$$c = w_1 - bw_2, \quad w_1 = \Phi(d_1), \quad w_2 = \Phi(d_2), \quad b = \exp(k - r\tau) > 0. \quad (18)$$

Under the money-market risk-neutral measure $\mathbb{Q}$ and the stock-numeraire measure $\mathbb{Q}^S$, respectively,

$$w_2 = \mathbb{Q}(S_T > K \mid \mathcal{F}_t) = \Phi(d_2), \quad w_1 = \mathbb{Q}^S(S_T > K \mid \mathcal{F}_t) = \Phi(d_1). \quad (19)$$

Here t is the valuation date, $T > t$ is the option maturity, S_T is the terminal underlying level, and $\mathcal{F}_t$ is the information set available at date t . Thus w_1 is the asset-leg digital probability, w_2 is the strike-leg digital probability, and b is the normalized discounted-strike prefactor. The explicit minus sign in (18) is part of the call decomposition. Equation (18) is the BSM specialization of the two-measure structure emphasized by DSE.

If $\hat{w}_1$ and $\hat{w}_2$ denote learned probability heads, the reconstructed value is

$$\hat{c}_{\text{dec}} = \hat{w}_1 - \exp(k - r\tau)\hat{w}_2. \quad (20)$$

A logistic output map constrains each learned weight to $(0, 1)$.

Proposition 2.2 (Two-leg price error bound). Let $c = w_1 - bw_2$ and $\hat{c} = \hat{w}_1 - b\hat{w}_2$ at the same state, with $b \geq 0$. Then

$$|c - \hat{c}| \leq |w_1 - \hat{w}_1| + b|w_2 - \hat{w}_2|. \quad (21)$$

Proof. Subtract the two representations and apply the triangle inequality.

A small reconstruction error does not by itself establish accurate local component motion. For a general two-leg representation $V = a_1w_1 + a_2w_2$ and scalar coordinate x ,

$$V_x = (a_1)_x w_1 + a_1(w_1)_x + (a_2)_x w_2 + a_2(w_2)_x. \quad (22)$$

Here a_1 and a_2 are differentiable scalar leg prefactors, w_1 and w_2 are differentiable component functions, and a subscript x denotes partial differentiation with respect to x . Consequently, decomposition accuracy and Greek accuracy are separate empirical questions.

2.4 Stochastic-discount extension

Under stochastic discounting, NDSE replaces the deterministic discounted-strike prefactor with bond-normalized and numeraire-matched terms. The spot and strike legs remain distinct, but their probability blocks are evaluated under different equivalent measures. The present empirical implementation remains in the analytic BSM environment so that structural effects can be separated from Monte Carlo and discretization noise. Measure-aware supervision is therefore a prospective extension rather than a component tested by the reported run.

3 Model hierarchy and training objective

3.1 Standardized supervised targets

Let

$$y = (c, \Delta, S\Gamma, \text{Vega}/S, \text{Vanna}, \text{Volga}/S) \quad (23)$$

collect the six supervised targets. Coordinate index $j \in \{1, \dots, 6\}$ follows the ordering in (23); observation index $i \in \{1, \dots, n_{\text{tr}}\}$ identifies a training example, and y_{ij} is its j th target coordinate. If the training sample contains n_{tr} observations, define the target location and scale by

$$m_j = \frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} y_{ij}, \quad s_j = \left[\frac{1}{n_{\text{tr}}} \sum_{i=1}^{n_{\text{tr}}} (y_{ij} - m_j)^2 \right]^{1/2} > 0. \quad (24)$$

Thus m_j is the training-sample mean and s_j is the training-sample population standard deviation of target coordinate j . This is the population-moment convention used by the implementation's standardization routine. The standardized target and prediction are

$$y_j^{\text{std}} = \frac{y_j - m_j}{s_j}, \quad \hat{y}_j^{\text{std}} = \frac{\hat{y}_j - m_j}{s_j}. \quad (25)$$

A hat denotes a model prediction. For Models B and C, $\hat{y}$ contains the direct price head and five direct Greek heads. For Model D, the supervised price coordinate is $\hat{c}_{\text{dec}}$ from (20), while the other five coordinates are the direct Greek heads. Thus the model-specific prediction vector used in (28) is

$$\hat{y}^{(B,C)} = (\hat{c}_{\text{direct}}, \hat{\Delta}, \hat{S\Gamma}, \widehat{\text{Vega}/S}, \widehat{\text{Vanna}}, \widehat{\text{Volga}/S}), \quad (26)$$

$$\hat{y}^{(D)} = (\hat{c}_{\text{dec}}, \hat{\Delta}, \hat{S\Gamma}, \widehat{\text{Vega}/S}, \widehat{\text{Vanna}}, \widehat{\text{Volga}/S}). \quad (27)$$

The common supervised loss is

$$\mathcal{L}_{\text{sup}} = \frac{1}{6} \sum_{j=1}^6 \lambda_j \mathbb{E} \left[\left(\hat{y}_j^{\text{std}} - y_j^{\text{std}} \right)^2 \right], \quad (28)$$

with $(\lambda_c, \lambda_{\Delta}, \lambda_{S\Gamma}, \lambda_{\text{Vega}/S}, \lambda_{\text{Vanna}}, \lambda_{\text{Volga}/S}) = (1, 1, 2, 1, 1, 1)$ in the reported run. The expectation in each training objective denotes the empirical mean over the observations on which that term is evaluated.

Model A has only a scalar normalized-price output. Its price loss, which is used below in (38), is

$$\mathcal{L}_c = \mathbb{E} \left[\left(\frac{\hat{c} - c}{s_c} \right)^2 \right] = \mathbb{E} \left[\left(\hat{c}^{\text{std}} - c^{\text{std}} \right)^2 \right]. \quad (29)$$

The four raw input coordinates are standardized by the same training-sample population-moment convention. Let

$$x = (k, \tau, r_{\tau}, \sigma_{\text{teacher}}), \quad x_i^{\text{std}} = \frac{x_i - \mu_i^x}{a_i^x}, \quad a_i^x > 0, \quad (30)$$

where $i \in \{1, \dots, 4\}$ indexes the coordinates in x , and μ_i^x and a_i^x denote their training-sample means and population standard deviations. Automatic differentiation is first performed with respect to x^{std} and converted back to raw-coordinate derivatives by

$$\frac{\partial c}{\partial x_i} = \frac{1}{a_i^x} \frac{\partial c}{\partial x_i^{\text{std}}}, \quad \frac{\partial^2 c}{\partial x_i \partial x_{\ell}} = \frac{1}{a_i^x a_{\ell}^x} \frac{\partial^2 c}{\partial x_i^{\text{std}} \partial x_{\ell}^{\text{std}}}. \quad (31)$$

In the second identity, $i, \ell \in \{1, \dots, 4\}$ may be equal or distinct. These scale factors are necessary for the raw-coordinate identities in (13).

3.2 Potential-consistency loss

Let the direct Greek heads be denoted by $\hat{q}$ and the corresponding quantities generated by automatic differentiation of a model-specific normalized-price potential c^{pot} by q^{pot} . For Models B and C, the diagnostic potential is the direct normalized-price head, $c^{\text{pot}} = \hat{c}_{\text{direct}}$. For Model D, it is the reconstructed two-leg price, $c^{\text{pot}} = \hat{c}_{\text{dec}}$; the direct price head is not differentiated for the Model D consistency term. Model B reports the resulting gaps only as diagnostics, whereas Models C and D include the following penalty:

$$\mathcal{L}_{\text{pot}} = \frac{1}{5} \sum_{q \in \{\Delta, S\Gamma, \text{Vega}/S, \text{Vanna}, \text{Volga}/S\}} \omega_q \mathbb{E} \left[\left(\frac{\hat{q} - q^{\text{pot}}}{s_q} \right)^2 \right], \quad (32)$$

where s_q is the corresponding training-sample scale and ω_q follows the same Greek weighting as (28). In the implementation, (32) is evaluated on a random 25% subset of each mini-batch to control the cost of higher-order automatic differentiation. The supervised term still uses every observation in the batch.

3.3 Decomposition and probability losses

Model D adds two learned probabilities and a direct price head. The price used for supervision and evaluation is the reconstructed value (20). The direct price head is retained as an internal diagnostic and is tied to the reconstruction by

$$\mathcal{L}_{\text{dec}} = \mathbb{E} \left[\left(\frac{\hat{c}_{\text{direct}} - \hat{c}_{\text{dec}}}{s_c} \right)^2 \right]. \quad (33)$$

The probability-head loss is

$$\mathcal{L}_W = \frac{1}{2} \mathbb{E} \left[(\hat{w}_1 - w_1)^2 + (\hat{w}_2 - w_2)^2 \right]. \quad (34)$$

The four objectives are

$$\mathcal{L}_A = \mathcal{L}_c, \quad (35)$$

$$\mathcal{L}_B = \mathcal{L}_{\text{sup}}, \quad (36)$$

$$\mathcal{L}_C = \mathcal{L}_{\text{sup}} + \lambda_{\text{pot}} \mathcal{L}_{\text{pot}}, \quad (37)$$

$$\mathcal{L}_D = \mathcal{L}_{\text{sup}} + \lambda_{\text{pot}} \mathcal{L}_{\text{pot}} + \lambda_{\text{dec}} \mathcal{L}_{\text{dec}} + \lambda_W \mathcal{L}_W. \quad (38)$$

The reported run uses $\lambda_{\text{pot}} = 1$, $\lambda_{\text{dec}} = 1$, and $\lambda_W = 0.5$.

3.4 Matched model definitions

All models use the same four input coordinates, three-layer hidden trunk, hidden width 64, SiLU activation, optimizer family, chronological samples, matched seeds, and validation rule. Their output layers necessarily differ: Models A, B, C, and D have 1, 6, 6, and 8 outputs, giving 8,705, 9,030, 9,030, and 9,160 trainable parameters, respectively. The design therefore matches the hidden architecture rather than imposing identical total parameter counts. No model receives a model-specific sampling distribution or local weighting scheme.

Table 1: Matched surrogate specifications

Model	Definition
A	Scalar normalized-price head. Delta, Gamma, Vega, Vanna, and Volga are recovered from automatic derivatives of the price potential.
B	Direct six-head model trained on price and the five selected sensitivity coordinates. No head-potential restriction is imposed.
C	Model B plus the potential-consistency loss (32).
D	Model C plus two supervised probability heads, reconstructed pricing, and the decomposition loss (33). The reported and supervised price is reconstructed from the two probability heads. The reported Greeks are direct heads, and their potential-consistency targets are derivatives of the reconstructed price rather than derivatives of the auxiliary direct price head.

3.5 Common validation rule

The validation criterion is the same standardized six-target RMSE for every model:

$$S_{\text{val}} = \frac{1}{\sum_{j=1}^6 \alpha_j} \sum_{j=1}^6 \alpha_j \frac{\text{RMSE}_j}{s_j}, \quad (\alpha_1, \dots, \alpha_6) = (1, 1, 1, 0.75, 0.75, 0.75). \quad (39)$$

Here RMSE_j is the validation-set root-mean-square error of target coordinate j in raw units, s_j is its training-sample scale from (24), and α_j is the fixed aggregation weight ordered as in (23). For Model A, every sensitivity entering (39) is produced from the scalar potential. For Models B–D, the reported sensitivity coordinates are the direct heads, while separate consistency diagnostics measure their agreement with the potential.

4 Data and empirical protocol

4.1 Date-causal market-state construction

The required local inputs are the S&P 500 close series from Stooq and five FRED series: DGS3MO, DGS2, DGS10, VIXCLS, and VXVCLS [13, 14, 15, 16]. The S&P 500 trading calendar is the master calendar. Treasury and volatility-index observations are merged by backward as-of joins, so a value used on date t has an observation date no later than t . An aligned observation more than ten calendar days old causes the run to fail. This procedure provides date-causal alignment of the downloaded histories; it does not reconstruct real-time vintage releases or eliminate later source revisions.

The common sample begins on December 4, 2007 because VXVCLS is the latest-starting required source. This truncates the historical window but does not differentially affect Models A–D, because every model uses the same aligned states. The final daily-state sample ends on March 18, 2026 and contains 4,600 dates. Source-age validation reports no violations.

The 30-day and 90-day implied-volatility proxies are

$$IV_{30} = \frac{\text{VIXCLS}}{100}, \quad IV_{90} = \frac{\text{VXVCLS}}{100}. \quad (40)$$

Let $\tau_{30} = 30/365$ and $\tau_{90} = 90/365$. For option maturity τ , the implied-volatility proxy is the affine interpolation or endpoint-segment extrapolation

$$IV(\tau) = \max \left\{ 10^{-8}, IV_{30} + \frac{\tau - \tau_{30}}{\tau_{90} - \tau_{30}} (IV_{90} - IV_{30}) \right\}. \quad (41)$$

The positivity floor is a numerical safeguard and is inactive for the validated source data. Let

$$g_t = \log \left(\frac{S_t}{S_{t-1}} \right), \quad \bar{g}_{20,t} = \frac{1}{20} \sum_{j=0}^{19} g_{t-j}. \quad (42)$$

Here t indexes dates on the master trading calendar, S_t is the S&P 500 close on date t , g_t is the one-period log return, j is the lag index, and $\bar{g}_{20,t}$ is the trailing twenty-observation mean return. Matching the implementation's rolling sample-standard-deviation convention, annualized twenty-day realized volatility is

$$RV_{20}(t) = \sqrt{252} \left[\frac{1}{19} \sum_{j=0}^{19} (g_{t-j} - \bar{g}_{20,t})^2 \right]^{1/2}. \quad (43)$$

The teacher volatility is

$$\sigma_{\text{teacher}}(\tau) = \alpha(\tau)IV(\tau) + (1 - \alpha(\tau))RV_{20}, \quad \alpha(\tau) = \begin{cases} 0.8, & \tau \leq 30/365, \\ 0.6, & \tau > 30/365, \end{cases} \quad (44)$$

where RV_{20} is the date- t quantity in (43). Define the three tenor-yield nodes (T_m, R_m) , $m = 1, 2, 3$, by

$$(T_m, R_m) \in \left\{ \left(0.25, \frac{\text{DGS3MO}}{100} \right), \left(2, \frac{\text{DGS2}}{100} \right), \left(10, \frac{\text{DGS10}}{100} \right) \right\}. \quad (45)$$

Here T_m is the tenor in years and R_m is the corresponding annualized yield in decimal form. The maturity-specific rate r_τ is the continuous piecewise-affine interpolant through these tenor-yield nodes, with the first or last affine segment used for extrapolation outside $[0.25, 10]$. The published Treasury series are constant-maturity yields quoted on an investment basis. The implementation divides them by 100 and uses the resulting annualized decimals directly as deterministic continuously compounded rate proxies; it does not bootstrap zero-coupon discount factors or convert the quotation convention. Consequently, r_τ is an empirical BSM input proxy rather than an arbitrage-exact zero curve.

The supervised network state is

$$(k, \tau, r_\tau, \sigma_{\text{teacher}}). \quad (46)$$

These four coordinates are sufficient for the analytic teacher and make the automatic sensitivity coordinates unambiguous.

4.2 Teacher panel and chronological splits

For each date, the teacher is evaluated on nine maturities,

$$\{3, 5, 7, 14, 21, 30, 45, 60, 90\} \text{ calendar days}, \quad (47)$$

and fifteen log-moneyness points,

$$\{-0.15, -0.10, -0.07, -0.05, -0.03, -0.02, -0.01, 0, 0.01, 0.02, 0.03, 0.05, 0.07, 0.10, 0.15\}. \quad (48)$$

Table 2: Chronological sample construction

Split	Dates	Observations	High-curvature observations	Date range
Train	2,537	342,495	136,998	2007-12-04 to 2017-12-29
Validation	1,008	136,080	54,432	2018-01-02 to 2021-12-31
Test	1,055	142,425	56,970	2022-01-03 to 2026-03-18

Thus each date contributes 135 contracts and the full panel contains 621,000 observations.

The high-curvature diagnostic slice is

$$|k| \leq 0.05, \quad \tau \leq 30/365. \quad (49)$$

Condition (49) is applied only after training. It does not change the training distribution, loss weights, mini-batches, or validation score.

4.3 Training controls

The completed extended-horizon run uses five matched seeds: 1234, 2234, 3234, 4234, and 5234. Teacher arithmetic, model parameters, and training tensors use float64. AdamW is used with learning rate 10^{-3} , weight decay 10^{-6} , mini-batch size 512, global gradient-norm clipping at 5, and deterministic seed handling when supported [11, 12]. The learning rate is controlled by a ReduceLROnPlateau scheduler with multiplicative factor 0.5, patience five, and relative improvement threshold 10^{-4} . Validation is performed after every epoch. For checkpoint selection and early stopping, a validation result counts as an improvement only when S_{val} falls by more than 10^{-5} ; training stops after 16 consecutive non-improving epochs. The safety cap is 960 epochs for Model A and 320 epochs for Models B–D. The higher Model A ceiling prevents its slower scalar-potential optimization from being truncated; it does not change the validation score or stopping rule. All twenty fitted models terminate through early stopping before their applicable cap.

4.4 Evaluation metrics

Global and high-curvature root-mean-square error (RMSE) and mean absolute error (MAE) are computed for

$$C, \quad \Delta, \quad \Gamma, \quad \text{Vega}, \quad \text{Vanna}, \quad \text{Volga}. \quad (50)$$

The normalized network outputs are mapped back to these reported coordinates by

$$\widehat{C} = S\widehat{C}, \quad \widehat{\Gamma} = \frac{\widehat{S\Gamma}}{S}, \quad \widehat{\text{Vega}} = S\widehat{\text{Vega}}/S, \quad \widehat{\text{Volga}} = S\widehat{\text{Volga}}/S, \quad (51)$$

with Delta and Vanna already in their reported units. The main-text tables emphasize RMSE, while the machine-readable output tables contain both RMSE and MAE. Models B–D also report the RMSE between each direct sensitivity head and the corresponding potential-generated quantity. Model D additionally reports the auxiliary direct-price versus reconstructed-price gap and the two probability-head errors.

Paired inference is based on within-seed RMSE improvements. Let s index a matched random seed, and let b and t denote the baseline and treatment models, respectively. Then

$$I_s = \text{RMSE}_{b,s} - \text{RMSE}_{t,s}. \quad (52)$$

Here I_s is the treatment improvement for seed s ; a positive value favors the treatment model. Accordingly, a label such as “B vs A” means treatment B against baseline A, while the numerical improvement is $\text{RMSE}_A - \text{RMSE}_B$. With $n_s = 5$ matched seeds,

$$\bar{I} = \frac{1}{n_s} \sum_{s=1}^{n_s} I_s, \quad s_I^2 = \frac{1}{n_s - 1} \sum_{s=1}^{n_s} (I_s - \bar{I})^2, \quad (53)$$

where $\bar{I}$ is the mean paired improvement and s_I is its sample standard deviation across seeds. The reported two-sided 95% interval is

$$\bar{I} \pm t_{n_s-1, 0.975} \frac{s_I}{\sqrt{n_s}}. \quad (54)$$

where $t_{\nu, p}$ denotes the p th quantile of a Student- t distribution with ν degrees of freedom. The two-sided p -value is obtained from the one-sample Student- t statistic $\bar{I}/(s_I/\sqrt{n_s})$ with $n_s - 1$ degrees of freedom.

4.5 Five-day hedging diagnostic

For every eligible test contract in the high-curvature slice, the model price and Delta initialize a discrete stock-cash hedge diagnostic. Let $t_0 < \dots < t_5$ be the six observed trading dates, let $S_i = S_{t_i}$, and let $\hat{C}_i$ and $\hat{\Delta}_i$ be the model price and Delta at t_i . Let d_i be the number of calendar days elapsed from t_0 to t_i , let θ_0 be the contract’s initial time to maturity, and define its remaining maturity by

$$\theta_i = \max \left\{ \theta_0 - \frac{d_i}{365}, 0 \right\}. \quad (55)$$

Matching the implementation, define $\rho_i := r_{\theta_i}(t_i)$ and use it as a right-endpoint constant-rate approximation over $(t_{i-1}, t_i]$. The resulting bookkeeping is self-financing conditional on this stated cash-accrual convention; because ρ_i is formed at the interval endpoint, it is a retrospective numerical diagnostic rather than an ex-ante trading rule. The initial cash account is

$$B_0 = \hat{C}_0 - \hat{\Delta}_0 S_0. \quad (56)$$

For $i = 1, \dots, 5$, with Δt_i the calendar-year fraction between t_{i-1} and t_i , the pre-rebalance portfolio value and the rebalanced cash account are

$$V_i = \hat{\Delta}_{i-1} S_i + B_{i-1} e^{\rho_i \Delta t_i}, \quad B_i = V_i - \hat{\Delta}_i S_i. \quad (57)$$

The reported hedge P&L is

$$\Pi = V_5 - C_{\text{teacher}}(t_5), \quad (58)$$

where $C_{\text{teacher}}(t_5)$ is the analytic teacher value of the same contract at the final diagnostic date. Thus Π is the terminal value of the model-initialized hedge net of that teacher value, and the final rebalancing identity in (57) is bookkeeping and does not alter Π . The experiment contains $n_h = 28,350$ valid contracts per model and seed. Within each seed, for hedge outcomes $\Pi_1, \dots, \Pi_{n_h}$, the implementation reports

$$\bar{\Pi} = \frac{1}{n_h} \sum_{j=1}^{n_h} \Pi_j, \quad \hat{\sigma}_{\Pi} = \left[\frac{1}{n_h} \sum_{j=1}^{n_h} (\Pi_j - \bar{\Pi})^2 \right]^{1/2}, \quad (59)$$

so the displayed P&L dispersion uses divisor n_h rather than the unbiased divisor $n_h - 1$. The remaining statistics are the empirical 95th percentile of $|\Pi|$ and the loss-tail statistic

$$\text{ES}_{0.95}^{\text{loss}} = \mathbb{E}[L \mid L \geq q_{0.95}(L)], \quad L = -\Pi, \quad (60)$$

where L is hedge loss and $q_{0.95}(L)$ is its empirical 95th percentile. The conditional expectation is implemented as the sample average over observations at or above that empirical loss quantile.

The exercise is a comparative numerical diagnostic rather than a trading strategy. It does not include transaction costs, bid–ask spreads, basis risk, parameter-estimation error, or market-option residuals.

5 Scientific-computing controls

5.1 Analytic and algebraic validation

The implementation validates the identities

$$c = w_1 - e^{k-r\tau}w_2, \quad (61)$$

$$\Delta = w_1, \quad (62)$$

$$\frac{\text{Vega}}{S} = S\sigma\tau\Gamma. \quad (63)$$

It also checks exact panel size, 135 contracts per date, positive state variables, finite targets, date uniqueness, date ordering, and the price-scale identity $C = Sc$. Every reported check passes. The maximum absolute error in the Vega–Gamma identity is 8.33×10^{-17} .

5.2 Stable floating-point evaluation

The teacher uses double precision. Normal probabilities are evaluated with a stable vectorized normal CDF, exponential arguments are clipped to the representable range of the selected floating type, and positive state variables are checked before labels are constructed. Non-finite training losses cause immediate failure rather than silent continuation.

5.3 Finite-difference audits

Automatic differentiation is the training mechanism, but an independent finite-difference audit tests the analytic teacher and the normalized potential identities at 32 randomly sampled panel states using seed 1234. For a scalar function f of an audited raw coordinate x , let $h > 0$ denote a coordinate perturbation. The fourth-order centered first derivative is

$$f'(x) \approx \frac{-f(x+2h) + 8f(x+h) - 8f(x-h) + f(x-2h)}{12h}, \quad (64)$$

and the fourth-order centered second derivative is

$$f''(x) \approx \frac{-f(x+2h) + 16f(x+h) - 30f(x) + 16f(x-h) - f(x-2h)}{12h^2}. \quad (65)$$

For two audited raw coordinates x and y , write $f_{xy} = \partial^2 f / (\partial x \partial y)$ and let $h_x > 0$ and $h_y > 0$ be their respective perturbations. The implementation uses the second-order centered stencil

$$f_{xy}(x, y) \approx \frac{f(x + h_x, y + h_y) - f(x + h_x, y - h_y) - f(x - h_x, y + h_y) + f(x - h_x, y - h_y)}{4h_x h_y}. \quad (66)$$

Following the truncation–roundoff balance in standard numerical analysis [5], the raw-coordinate step rules are

$$h_1(x) = \max \left\{ h_{1,\text{requested}}, \varepsilon_{\text{mach}}^{1/5} \max(1, |x|) \right\}, \quad (67)$$

$$h_2(x) = \max \left\{ h_{2,\text{requested}}, \varepsilon_{\text{mach}}^{1/6} \max(1, |x|) \right\}, \quad (68)$$

$$h_{\text{mix}}(x) = \max \left\{ h_{\text{mix},\text{requested}}, \varepsilon_{\text{mach}}^{1/4} \max(1, |x|) \right\}. \quad (69)$$

Here $\varepsilon_{\text{mach}}$ is machine epsilon for the float64 arithmetic used in the audit. The symbols $h_{1,\text{requested}}$, $h_{2,\text{requested}}$, and $h_{\text{mix},\text{requested}}$ are the nominal step floors, equal to 10^{-4} , 10^{-3} , and 10^{-4} in the reported implementation. The resulting h_1 , h_2 , and h_{mix} are used for the first-, second-, and centered mixed-derivative audits, respectively; for a mixed derivative, the h_{mix} rule is applied separately to each participating coordinate. Large relative errors can occur when the analytic target is extremely close to zero; the corresponding absolute errors remain small and are therefore reported jointly.

5.4 Smooth automatic differentiation

Automatic differentiation evaluates derivatives of programmatically specified numerical functions without replacing them by finite-difference approximations [8]. A second-order consistency objective requires nontrivial second automatic derivatives. A piecewise-linear ReLU network has zero classical second derivative almost everywhere, which is poorly matched to Gamma, Vanna, and Volga identities generated from a scalar price potential. Every hidden layer therefore uses the smooth SiLU activation. This is a structural requirement rather than a cosmetic architecture choice.

6 Empirical results

6.1 Global and high-curvature accuracy

Table 3 reports mean test-set RMSE across the five matched seeds. Boldface identifies the lowest mean in each column. These entries are descriptive summaries; the subsequent Q1–Q3 subsections use matched-seed differences, confidence intervals, and p -values to assess the relevant model comparisons.

Table 3: Mean global test-set RMSE across five matched seeds

Model	Price	Delta	Gamma	Vega	Vanna	Volga
A	0.319084	0.003180	5.548×10^{-5}	5.677856	0.052545	125.717977
B	2.015389	0.003734	1.652×10^{-5}	3.288375	0.015453	29.691537
C	1.012270	0.003467	2.157×10^{-5}	3.774917	0.022052	41.388709
D	0.377107	0.005130	3.128×10^{-5}	5.789156	0.037245	61.915029

Table 4 reports the same descriptive metrics on the ex-post high-curvature slice. The broad mean ranking resembles the global table, but the magnitude and statistical resolution of selected differences change in this more curvature-intensive region; those paired comparisons are reported below.

Table 4: Mean high-curvature-slice RMSE across five matched seeds

Model	Price	Delta	Gamma	Vega	Vanna	Volga
A	0.273984	0.002967	6.940×10^{-5}	5.320263	0.067086	119.700088
B	1.782842	0.003569	2.143×10^{-5}	2.948049	0.016961	27.124095
C	0.579231	0.003368	3.018×10^{-5}	2.954409	0.026369	34.166614
D	0.358892	0.005131	4.275×10^{-5}	5.005248	0.048634	59.210957

6.2 Q1: direct Greek supervision

Model A versus Model B isolates direct supervision on the five sensitivity targets. Relative to Model A, Model B lowers global Gamma, Vega, Vanna, and Volga RMSE by approximately 70.2%, 42.1%, 70.6%, and 76.4%, respectively. Each improvement occurs in all five seeds and is statistically resolved at the 95% level. By contrast, global Delta RMSE rises by approximately 17.4%, from 0.003180 to 0.003734; the paired interval includes zero and $p = 0.3364$. On the high-curvature slice, Delta RMSE also rises, from 0.002967 to 0.003569, with $p = 0.2036$.

The higher-order Greek gains are accompanied by a price tradeoff. Global price RMSE rises from 0.319 to 2.015; the matched mean improvement is -1.696 with 95% confidence interval $[-2.355, -1.037]$ and two-sided $p = 0.0020$. The high-curvature price comparison also significantly favors Model A. On the high-curvature slice, the Gamma, Vanna, and Volga gains remain statistically resolved, whereas the Vega gain is directional but does not cross the 5% threshold ($p = 0.0644$). The completed stopping and cap-binding audit therefore sharpens the Q1 interpretation: direct Greek supervision provides large incremental value for higher-order sensitivities, but not for Delta, and it imposes a substantial price-fit cost under the reported matched hidden architecture and loss weights.

Table 5: Selected paired RMSE improvements; positive values favor the treatment model

Slice	Quantity	Treatment vs baseline	Mean improvement	95% CI	p -value
Global	Price	B vs A	-1.696305	$[-2.355141, -1.037469]$	0.0020
Global	Delta	B vs A	-0.000554	$[-0.001963, 0.000855]$	0.3364
Global	Gamma	B vs A	3.896×10^{-5}	$[2.556, 5.236] \times 10^{-5}$	0.0013
Global	Vega	B vs A	2.389481	$[0.758526, 4.020436]$	0.0153
Global	Vanna	B vs A	0.037092	$[0.021460, 0.052724]$	0.0027
Global	Volga	B vs A	96.026440	$[56.003580, 136.049300]$	0.0026
High curvature	Vega	B vs A	2.372214	$[-0.227475, 4.971904]$	0.0644
Global	Price	C vs B	1.003120	$[0.508565, 1.497674]$	0.0049
High curvature	Price	C vs B	1.203610	$[0.472943, 1.934278]$	0.0102
Global	Price	D vs C	0.635162	$[0.237866, 1.032458]$	0.0113
Global	Price	D vs A	-0.058023	$[-0.109412, -0.006634]$	0.0350
High curvature	Price	D vs A	-0.084908	$[-0.193826, 0.024011]$	0.0964

6.3 Q2: potential consistency

Model B versus Model C isolates the head-potential consistency restriction. The primary effect is not a uniform reduction in standalone Greek RMSE. Instead, the restriction sharply reduces disagreement between the direct heads and the derivatives of the common pricing potential.

Every listed gap is smaller for Model C in each of the five matched seeds. Model C also lowers global price RMSE by 49.8% relative to Model B and high-curvature price RMSE by 67.5%; both paired improvements are statistically resolved. Global Delta improves by 7.1% relative to Model B, but the

Table 6: Mean head-potential gap RMSE and reduction from Model B to Model C

Quantity	Model B	Model C	Reduction
Delta	0.019371	0.003605	81.4%
Spot-scaled Gamma	2.924839	0.104413	96.4%
Normalized Vega	0.006388	0.000709	88.9%
Vanna	0.582214	0.019332	96.7%
Normalized Volga	0.255277	0.006995	97.3%

paired comparison is not significant and Model A retains the lowest mean Delta RMSE. Gamma, Vega, and Vanna worsen directionally without reaching the 5% level, while global Volga rises from 29.69 to 41.39 and the difference is statistically resolved at $p = 0.0295$. The empirical interpretation is therefore specific: potential consistency produces a much more coherent pricing-and-risk map and recovers much of the price sacrifice of direct Greek supervision, but it remains a separate objective from pointwise Greek minimization.

6.4 Q3: two-leg decomposition

Model D versus Model C isolates the supervised two-leg representation. Model D lowers global price RMSE from 1.012 to 0.377, a 62.7% reduction. The matched price improvement is positive in all five seeds, with mean 0.635, 95% confidence interval $[0.238, 1.032]$, and $p = 0.0113$. On the high-curvature slice, the mean price RMSE falls from 0.579 to 0.359, but the paired confidence interval includes zero and $p = 0.174$. Relative to Model A, however, Model D has an 18.2% higher mean global price RMSE. The paired A–D comparison favors Model A, with a mean A-minus-D improvement of -0.058 , 95% confidence interval $[-0.109, -0.0066]$, and $p = 0.0350$. On the high-curvature slice, Model A also has the lower mean price RMSE, although that paired difference is not statistically resolved ($p = 0.0964$).

The decomposition itself is learned accurately. The mean auxiliary normalized-price gap RMSE between $\hat{c}_{\text{direct}}$ and $\hat{c}_{\text{dec}}$ is 3.41×10^{-4} . The asset-leg and strike-leg probability-head RMSEs are 0.00916 and 0.00908, respectively.

The global price gain relative to Model C is accompanied by weaker direct Greeks. Model D raises global Delta, Gamma, Vega, Vanna, and Volga RMSE by approximately 47.9%, 45.0%, 53.4%, 68.9%, and 49.6%, respectively. The paired differences are statistically resolved for Delta, Vega, and Volga. Gamma is directional but not resolved at the 5% level, and Vanna is marginal with $p = 0.0512$. On the high-curvature slice, the deterioration is statistically resolved for Delta, Vega, Vanna, and Volga, but not Gamma.

Table 7: Model D versus Model C: paired global RMSE improvements

Quantity	Mean improvement	95% CI	p -value
Price	0.635162	$[0.237866, 1.032458]$	0.0113
Delta	-0.001663	$[-0.002514, -0.000811]$	0.0056
Gamma	-9.711×10^{-6}	$[-2.090 \times 10^{-5}, 1.482 \times 10^{-6}]$	0.0736
Vega	-2.014239	$[-3.447302, -0.581176]$	0.0175
Vanna	-0.015193	$[-0.030510, 0.000124]$	0.0512
Volga	-20.526320	$[-38.082530, -2.970110]$	0.0315

Accordingly, Model D supports decomposition fidelity and price repair relative to Model C, not

uniform sensitivity or price dominance under the shared hidden-trunk specification and loss weights.

6.5 Hedging diagnostics

The five-day hedging results are economically close across the four models.

Table 8: Mean five-day hedging diagnostics across matched seeds

Model	Mean P&L	P&L SD	$ \text{P\&L} _{0.95}$	ES-style loss	Contracts
A	3.4994	17.3880	30.9413	45.3128	28,350
B	4.1239	17.4680	31.0137	44.8526	28,350
C	3.8390	17.3613	30.9600	45.0619	28,350
D	3.1257	17.3848	30.8360	45.7420	28,350

Model C has the smallest P&L standard deviation, Model D has the smallest mean P&L and 95% absolute-P&L quantile, and Model B has the lowest expected-shortfall-style loss. The maximum differences in P&L standard deviation and the 95% absolute quantile are both below 0.7% of their respective cross-model levels. No paired hedging inference is produced by the implementation. These differences do not support a claim that structural supervision produces clear hedging dominance.

6.6 Stopping and cap-binding status of the completed run

The completed benchmark uses a 960-epoch safety cap for Model A and a 320-epoch cap for Models B–D. All twenty fits terminate through the common early-stopping rule. The five Model A fits stop at epochs 659, 677, 578, 496, and 695, with best checkpoints at epochs 643, 661, 562, 480, and 679. None reaches the applicable cap, and no selected Model A checkpoint lies within the final ten epochs of its run.

Table 9: Stopping and cap-binding summary across five matched seeds

Model	Safety cap	Mean final epoch	Mean best epoch	Early stops	Cap hits
A	960	621.0	605.0	5	0
B	320	202.4	186.4	5	0
C	320	245.4	229.4	5	0
D	320	257.0	241.0	5	0

Within the reported experiment, the extended horizon resolves the earlier cap-binding issue and places all models under the same stopping rule, validation score, patience, optimizer family, hidden architecture, and loss-weight specification; it does not establish global optimization convergence. The model-specific ceilings are safety limits rather than preferential model-selection rules: every selected checkpoint precedes its cap, and every fit is stopped by the same patience criterion. Model A simply requires a longer optimization path to satisfy that criterion. The A–B comparisons can therefore be interpreted without the previous epoch-cap caveat.

7 Interpretation

The experiment separates three interventions that are often combined in derivative-surrogate design: adding derivative targets, requiring agreement with a common pricing potential, and supervising economically interpretable decomposition blocks. Each intervention changes both the feasible representation and the allocation of a shared network’s capacity.

The A–B comparison shows that additional derivative labels can redirect accuracy from the scalar price toward selected higher-order risks. The B–C comparison shows that integrability can improve sharply without producing the lowest standalone error for every Greek. The C–D comparison shows that interpretable intermediate targets can repair price error relative to a constrained multi-output model while competing with direct sensitivity objectives. These patterns are therefore consistent with a multi-objective approximation problem rather than contradictory model rankings.

For practical model selection, the primary use case determines which errors and structural residuals deserve the greatest weight. Price, direct-Greek, potential-consistency, decomposition, and hedge diagnostics should be monitored jointly rather than collapsed into a single complexity ranking. The controlled BSM experiment provides evidence for this task-specific view, but it does not by itself establish a production ranking for broader pricing models or market environments.

8 Limitations

Several limitations qualify the empirical evidence.

First, the teacher is analytic BSM rather than a market-option panel. The experiment measures approximation and structural fidelity under controlled labels; it does not show that the surrogate corrects model misspecification or market microstructure effects.

Second, the market series provide realistic time variation in spot, rates, realized volatility, and implied-volatility proxies, but the teacher fixes dividend yield at zero. The downloaded FRED histories are date-causally aligned rather than vintage-reconstructed, and the constant-maturity Treasury yields are used directly as continuous-rate proxies rather than being converted into an arbitrage-consistent zero curve. The S&P 500 is also an index rather than a directly tradable hedge instrument. A production hedging study would require an explicit futures or ETF hedge, basis treatment, transaction costs, dividends or forwards, bid–ask spreads, and an ex-ante cash-accrual convention. The present fixed-rule diagnostic is therefore distinct from end-to-end policy-learning approaches that optimize hedging under market frictions [9].

Third, VVCLS determines the common start date of December 4, 2007. The sample contains the global financial crisis, the low-rate period, the pandemic, and the subsequent rate transition, but it does not test earlier regimes. The truncated start affects external historical coverage, not the fairness of the matched A–D comparison.

Fourth, five seeds are sufficient to reveal large and stable differences but provide limited power for smaller effects. Confidence intervals should therefore be interpreted as matched-seed evidence rather than as asymptotic inference from a large independent sample.

Fifth, all twenty fits satisfy the reported early-stopping rule, but the resulting optimization and model-selection status remains conditional on the selected AdamW schedule, matched hidden architecture, model-specific output heads, loss weights, validation score, and patience parameter.

Model A requires a materially longer optimization path than Models B–D, which shows that convergence speed differs across objectives. Alternative architectures or multi-objective weighting schemes could move the empirical Pareto frontier even though the present epoch-cap concern has been resolved.

Sixth, the BSM two-leg decomposition is exact and low dimensional. More general deterministic-discount engines require conditional-law or simulation-based blocks, while stochastic-discount models require numeraire-matched labels. Those extensions will add teacher error and numerical noise that are absent from the present experiment.

Seventh, the five-day hedge is a controlled numerical diagnostic. The implementation does not produce paired hedging inference, and the cross-model differences are small. The hedge results should not be interpreted as evidence of trading profitability or production robustness.

Finally, consistency restrictions enlarge the set of simultaneous requirements and therefore need not improve every standalone metric. The current loss weights and hidden-trunk size are not claimed to be Pareto optimal.

9 Conclusion

This paper develops a derivative-surrogate hierarchy that combines analytic price and Greek labels with two forms of structural supervision: agreement with a normalized pricing potential and reconstruction from supervised probability legs. The matched-seed experiment indicates that these mechanisms address different approximation failures rather than forming a uniformly improving sequence.

Direct Greek labels improve selected higher-order sensitivities at a price-fit cost; potential penalties improve internal coherence; and two-leg supervision adds interpretable decomposition fidelity and repairs price error relative to Model C without dominating all price, Greek, or hedge criteria. Because every fit stops before its applicable safety cap, the reported tradeoffs are not attributable to cap binding within this experiment. The conclusions remain specific to the controlled BSM teacher and support task-dependent surrogate design rather than a universal architecture ranking.

Appendix

A Implementation outputs and reproducibility

The empirical implementation writes the following principal artifacts:

- date-causally aligned daily states and the full teacher panel;
- source-coverage, data-validation, and finite-difference audit tables;
- twenty fitted model files and twenty epoch-history files;
- per-seed and summary accuracy metrics;
- paired RMSE inference across matched seeds;

- head-potential and decomposition-consistency diagnostics;
- per-seed and summary hedging diagnostics; and
- a manifest containing data paths, dtypes, seeds, model definitions, hyperparameters, and execution controls.

The reported run uses the local files

spx_stooq.csv, DGS3M0.csv, DGS2.csv, DGS10.csv,
VIXCLS.csv, and VVCLS.csv.

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